CCC-C-421a
November 13, 1963
SUPERSEDING
Fed. Spec. CCC-C-421
April 12, 1961

FEDERAL SPECIFICATION

CLOTH, DENIM, COTTON, SHRUNK AND UNSHRUNK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, cotton, denim.

1.2 Classification.

1.2.1 *Types, classes, and styles.* The cloth shall be of the following types, classes, and styles (table I, 6.2 and 6.5):

Type I.—8.04 to 8.78 ounces.
Type II.—8.95 to 9.75 ounces.
Type III.—9.85 to 10.97 ounces.
Type IV.—11.06 to 12.30 ounces.
Class 1.—Unshrunk.
Class 2.—Shrunk.
Style A.—Whiteback.
Style B.—Colored.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).
PPP-B-636—Box, Fiberboard.
CCC-T-191—Textile Test Methods.
PPP-P-51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Standard sample.** When a standard sample is available, the cloth shall match the sample for shade and shall be equal to, or better than, the sample with respect to all characteristics for which the sample is referenced (6.3).

3.2 **Material.** The cotton shall be thoroughly cleaned, carded, drawn, and evenly spun into singles yarn.

3.3 **Color.** The color shall be as specified and shall match the standard sample when available (3.1, 6.2 and 6.3). Style A, whiteback cloth shall contain dyed warp yarns and white or tinted filling yarns. When Blue 3301 shade is specified (6.2 and 6.6), the shade shall be obtained by the use of vat dyestuffs. The filling yarns may be either blended or tinted. Style B cloth shall contain dyed warp and filling yarns.

3.3.1 *Color matching.* The color shall match the available standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* Unless otherwise specified (6.2), the dyed cloth shall show colorfastness to light (after 40 hours exposure), water, laundering, and crocking equal to, or better than, the standard sample when tested as specified in 4.4. When no standard sample is available, or when the standard sample is not a reference for colorfastness, the dyed cloth shall show good colorfastness to light (after 40 hours exposure), water, laundering and fair colorfastness to crocking when tested as specified in 4.4.

3.4 **Physical requirements.** The finished cloth shall conform to the requirements shown in table I for the applicable type and class when tested as specified in 4.4.

TABLE I.—*Physical requirements*

Type	Class	Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength, grab method (minimum)	
			Warp	Filling	Warp	Filling
		Ounces			*Pounds*	*Pounds*
I	1	8.04	62	39	145	53
	2	8.78	63	42	145	58
II	1	8.95	66	43	150	65
	2	9.75	67	46	150	70
III	1	9.85	68	42	167	70
	2	10.97	69	45	167	75
IV	1	11.06	70	42	177	85
	2	12.30	71	45	177	90

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3.5 Weave. Unless otherwise specified, the weave shall be a 3-harness construction.

$\frac{2}{1}$ Right-hand twill (6.2).

3.5.1 *Width.* The width exclusive of selvages shall be as specified (6.2).

3.6 Nonfibrous material. The total nonfibrous material content of type I cloth shall not exceed 14 percent and for types II, III, and IV cloths shall not exceed 13 percent, when tested as specified in 4.4

3.7 Shrinkage or stretch. Class 2 cloth shall be preshrunk and shall not shrink more than 2 percent or stretch more than 0.75 percent in either direction of the warp or filling when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark, either on the cloth or on the ticket or package.

3.8 pH. The pH of the cloth shall be not lower than 8.0 nor more than 10.0 when tested as specified in 4.4.

3.9 Length. Unless otherwise specified (6.2), the cloth shall be furnished in pieces of not less than 50 continuous yards or more than 100 continuous yards.

3.10 Fiber identification. Each piece, roll, or bolt of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.11 Workmanship. The cloth shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the limits set by the applicable acceptable quality levels or point levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Except when otherwise specified by Civil Government agencies, inspection shall be in accordance with the provisions of MIL-STD-105 and 4.2.1 through 4.2.3.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece, roll, or bolt shall be examined on the face side and all defects as defined in Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except that spots, stains, and streaks shall be counted regardless of cause (i.e., not restricted to rust, oil, or grease) and only those slubs and knots which exceed the maximum acceptable limits shown for each shall be scored (4.2.1.2). Defects on the selvage shall not be scored unless the body of the cloth is affected. No yard shall be penalized more than four points. For purposes of this examination, a yard shall begin at the last scored defect. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 pieces, rolls, or bolts taken from 25 containers. An approximately equal number of yards shall be examined in each piece, roll or bolt of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 40.00. Point computation shall be as follows:

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$$\frac{T \times 3600}{750 \times C} = P$$

where:
T = Total points scored in sample size.
C = Contracted width of material (inches).
P = Points per 100 square yards.

4.2.1.1 *Demerit Points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension —one point

For defects exceeding 3 inches, but not exceeding 6 inches in any dimension—two points

For defects exceeding 6 inches, but not exceeding 9 inches in any dimension—three points

For defects exceeding 9 inches in any dimension—four points

The following defects when present, shall be scored four points for each yard in which they occur:

Defects

Objectionable odor or not clean throughout.

Poor dye penetration—cloudy, mottled, or streaky throughout.

Uneven weaving throughout.

Selvage rolled, folded, corded, or slack, continuously or intermittently throughout the piece.

Width less than specified.

Preshrinking process or trademark identified on the cloth.

Filling yarn for style A not white or tinted as specified.

4.2.1.2 *Limits of slubs and knots.* Only slubs and knots which exceed the limits shown in figure 1 shall be scored.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces, rolls, or bolts.* During the yard-by-yard examination, each piece, roll, or bolt in the sample shall be examined for gross length. Any gross length found to be less than the minimum specified or more than two yards below the gross length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces, rolls, or bolts in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of pieces, rolls, or bolts in the sample is less than the total of the gross lengths marked on the ticket.

4.2.3 *Examination for shade.* During the yard-by-yard examination each piece, roll, or bolt in the sample shall be examined for shade. Any piece, roll, or bolt in the sample off shade (compared to the standard sample and applicable shade range) shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 *Examination for identification of preshrinkage process and noncompliance with Textile Fiber Products Identification Act.* During the yard-by-yard examination each piece, roll, or bolt in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L4 and the AQL shall be 4.0 defects per 100 units of MIL-STD-105.

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Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
	Preshrinkage process identified by name or trademark on container.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: Incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross weight exceeds requirement.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 3-1/2 continuous yards, full width of cloth. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified.

TABLE II.—*Test Methods*

Characteristic	Requirement reference	Test method no.
Colorfastness to:		
Light[1]	3.3.2	5660
Water	3.3.2	5630
Laundering	3.3.2	5610
Crocking	3.3.2	5651
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking Strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave[2]	3.4	Visual
Non fibrous material[3]	3.6	2611
Shrinkage or stretch:[4]		
Warp	3.7	5550
Filling	3.7	5550
Acidity (pH)	3.8	2811

[1] In the event of a dispute resulting from the test method 5660, as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[2] One determination shall be made from each unit of product and the result shall be reported as "pass or fail."

[3] The non-fibrous materials content shall be determined with the following formula:

$$\text{Nonfibrous-materials content} = \frac{O\text{-}S}{0} \times 100$$

[4] The three specimens for this test shall be selected from the cloth (unit of product) as follows: One specimen from each side of the cloth within 3 inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns. Test results shall be reported as stretch or shrinkage.

5. PREPARATION FOR DELIVERY

(For civil agency procurements, Fed. Std. No. 102 should be referred to for definitions and application of the various levels of packaging and packing.)

5.1 Packaging. Packaging shall be level A, B, or C, as specified (6.2).

5.1.1 *Levels A and* C. The cloth shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

5.1.2 *Level B, standard pack (civil agencies).* Approximately 50 continuous yards (plus or minus 5 percent, of cloth of the type, class, style, and color specified shall be put-up as flat-folded pieces (bolts) in accordance with the requirements of level A, put-up, of PPP-P-51 or, at the option of the contractor, the cloth may be put-up by winding on a flat chipboard or cardboard insert of not less than 1/8 inch thick by 12 ± 1/4 inches wide by necessary length.

Bolts put-up as above, shall be completely wrapped in kraft paper, 30 pounds (minimum) basis weight (24 by 36—500), conforming to grade B of UU-P-268. The bolt shall be so wrapped that the paper shall encircle the bolt twice, with width of paper sufficient to fold over and protect the ends

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of the bolt. Wrapping shall be secured by gummed paper tape 2-1/2 inches minimum width, conforming to UU-T-111, class 3, applied over the folded ends of wrapping, centered approximately to the width of the bolt. Wrapping shall be applied in a manner which shall permit the bolt identification tag attached to selvage of material to be on the outside of the wrapped bolt.

5.2 Packing. Packing shall be level A, B, or C, as specified (6.2).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51. When level B is specified by civil agencies for cloth approximately 50 yards in length, the cloth shall be packed in accordance with 5.2.2.

5.2.2 *Level B, standard pack (civil agencies).* Bolts put-up and wrapped, as specified in 5.1.2, shall be packed in new fiberboard containers conforming to type I or II, class 1, style FTC, FOL, SFF, or RSC-L, of PPP-B-636, except that in no event shall the gross weight of box and contents exceed 90 pounds, nor the burst test of the fiberboard be less than 275 p.s.i. Sleeve liner for type RSC-L boxes shall be of not less than 175 p.s.i., top and bottom full width by full length fiberboard pads shall test not less than 100 p.s.i. Box closure shall be in accordance with appendix of the box specification. According to weight of wrapped bolts, 12 bolts of up to 6 pounds each, 8 bolts of up to 10 pounds each, 6 bolts of from 11 to 14 pounds each, 4 bolts of from 15 to 21 pounds each, 3 bolts of from 22 to 28 pounds each, 2 bolts of from 29 to 42 pounds each, or one bolt of over 43 pounds shall be packed per shipping container. Containers of 6 to 12 bolts shall be packed with bolts in two stacks side by side, 4 bolts or less shall be in a single stack within the container.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-51. When level B packaging and packing are required by civil agencies as specified in 5.1.2 and 5.2.2, marking shall be in accordance with 5.3.1.

5.3.1 *Civil agencies (level B).* Each bolt shall be provided with the identification tag on the outside of the wrapping. No other marking required for the individual bolts. Each shipping container shall be marked in accordance with Fed. Std. No. 123 and with the additional precautionary markings as follows:

On each side or side panel: "DO NOT STAND ON END in 1-inch letters.

For quantity contained: ".....BOLTS - Total yardage .....YDS." indicating total number of bolts and total yardage contained.

6. NOTES

6.1 Intended Use. The cloth covered by this specification is intended for use in the manufacture of clothing items.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and style required (1.2.1).
(*c*) Color of cloth required (3.3 and 6.6).
(*d*) When colorfastness required is other than specified (3.3.2).
(*e*) When weave required is other than specified (3.5).
(*f*) Width required (3.5.1).
(*g*) When length is other than specified (3.9 and 6.4).
(*h*) Selection of applicable levels of packaging and packing (5.1, 5.2 and 6.4).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, alternatives, as indicated in PPP-P-51, may be exercised in the preparation for delivery of cloth.

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6.3 Standard sample. For access to a standard sample, address the procuring office issuing the invitation for bids.

6.4 The standard pack requirements in paragraphs 5.1.2 and 5.2.2 cover packaging and packing for bolts of cloth approximately 50 continuous yards in length and are intended for use in procurements of Stores stock replenishments. Procuring officers should use the Standard Pack requirements when it is known that the cloth approximately 50 continuous yards in length will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user. For civil agency procurement of cloth other than approximately 50 continuous yards in length under level B requirements, the level A packaging and level B packing specified in PPP-P-51 shall be used.

6.5 Transportation description. Transportation description and maximum weights applicable to this commodity are:

Rail:

Cotton piece goods, not otherwise indexed by name, in the original piece.
Carload minimum weight—any quantity.

Motor:

Cotton piece goods, not otherwise indexed, in the original piece.
Motor volume minimum weight—any quantity.

6.6 Note: A suggested but not mandatory dyestuff formulation for Blue 3301 is the use of Vat Blue 43 for the warp yarns and Denivat Everblack-W blended with natural cotton for the filling yarns.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

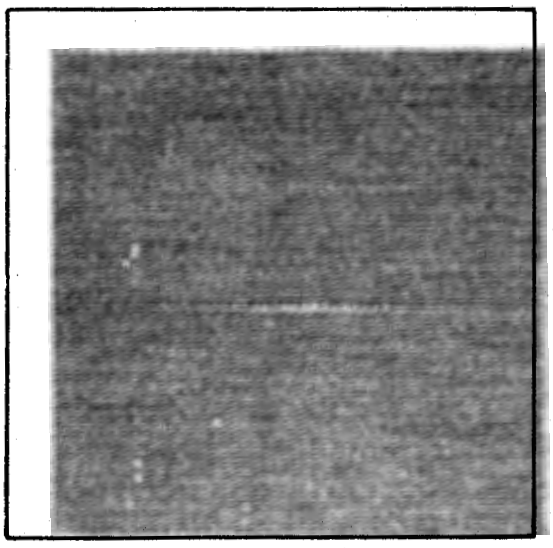
Army—Q

Navy—S

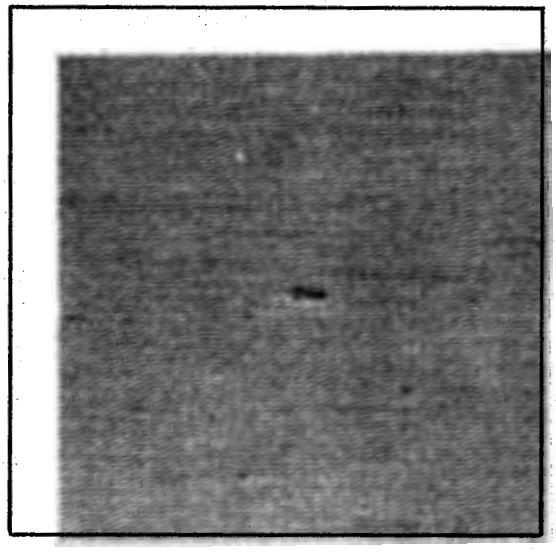
Air Force—ASD

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Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

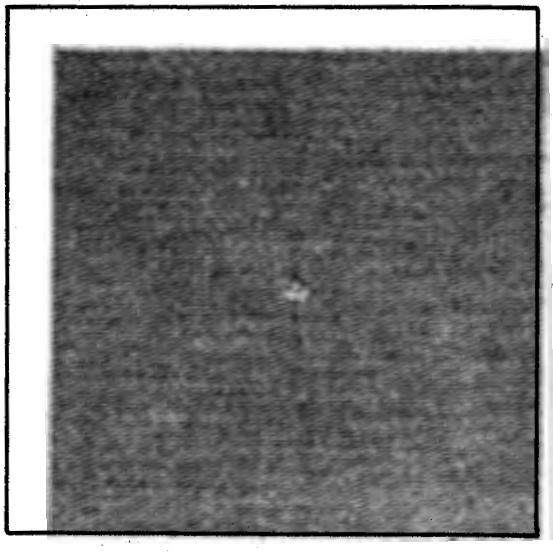


FIGURE 1. Limits of acceptability.

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